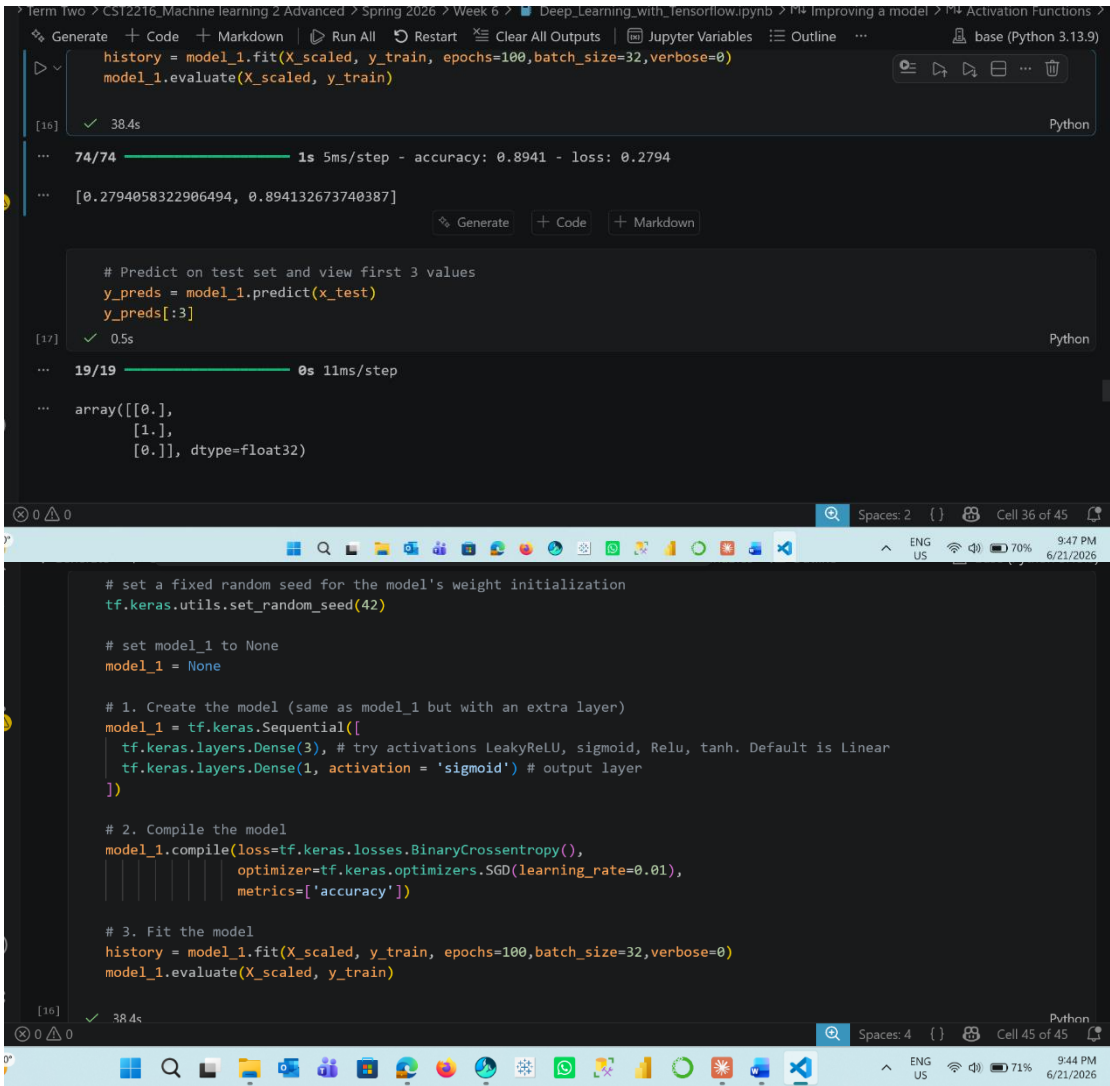


CST2216 Machine learning Lab_5_Work
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best model hyper parameters achieved in a table.

Metric name/Evaluation method	Defintion	Code
Accuracy	Out of 100 predictions, how many does your model get correct? E.g. 95% accuracy means it gets 95/100 predictions correct.	<code>sklearn.metrics.accuracy_score()</code> or <code>tf.keras.metrics.Accuracy()</code>



The image displays two screenshots of a Jupyter Notebook interface. The top screenshot shows the execution of a Keras model training and evaluation process. The code includes `model_1.fit(X_scaled, y_train, epochs=100, batch_size=32, verbose=0)` and `model_1.evaluate(X_scaled, y_train)`. The output shows a progress bar at 74/74, a time of 1s 5ms/step, an accuracy of 0.8941, and a loss of 0.2794. The bottom screenshot shows the creation and compilation of a new model, `model_1`, with a Sequential architecture consisting of a Dense(3) layer and a Dense(1) layer with a sigmoid activation. The model is compiled with BinaryCrossentropy loss, SGD optimizer, and accuracy metric. The code also includes a fixed random seed for weight initialization and a fit method call.

```
# set a fixed random seed for the model's weight initialization
tf.keras.utils.set_random_seed(42)

# set model_1 to None
model_1 = None

# 1. Create the model (same as model_1 but with an extra layer)
model_1 = tf.keras.Sequential([
    tf.keras.layers.Dense(3), # try activations LeakyReLU, sigmoid, Relu, tanh. Default is Linear
    tf.keras.layers.Dense(1, activation = 'sigmoid') # output layer
])

# 2. Compile the model
model_1.compile(loss=tf.keras.losses.BinaryCrossentropy(),
               optimizer=tf.keras.optimizers.SGD(learning_rate=0.01),
               metrics=['accuracy'])

# 3. Fit the model
history = model_1.fit(X_scaled, y_train, epochs=100, batch_size=32, verbose=0)
model_1.evaluate(X_scaled, y_train)
```

Note: The loss function going down means the model is improving (the predictions it's making are getting closer to the ground truth labels).

